Smart Control and Monitoring System for the College of Computing and Information Sciences (CoCIS), Makerere University

PROGRAMMES: BSE III & CS II

ACADEMIC YEAR: 2026/2027

1. Background and Problem Statement

The College of Computing and Information Sciences (CoCIS) at Makerere University houses valuable equipment that requires reliable security measures to prevent unauthorized access and ensure the safety of institutional resources. In addition to security concerns, the college faces high electricity costs, partly due to lights being left on in unoccupied rooms.

During a management meeting held at the beginning of October 2026, it was agreed that an automated system should be developed to address these challenges. The proposed solution will initially be implemented as a pilot project covering two lecture rooms and the main entrance of the building.

2. System Requirements and Proposed Solution

The system will use ESP32 microcontrollers to automate access control, monitor building occupancy, and manage lighting in the selected lecture rooms.

2.1. Access Control and Occupancy Monitoring

The main entrance will be divided into two designated sections: one for people entering the building and another for those exiting. Every person accessing the building will be identified using a Radio Frequency Identification (RFID) tag, which must be scanned at both entry and exit points.

When a valid RFID tag is detected, the system will verify the user's identity and grant access by activating the appropriate door mechanism. The system will also maintain records of entry and exit events, enabling management to determine who is inside the building at any given time.

2.2. Automated Lighting Control

The two selected lecture rooms will be equipped with an automated lighting control system. The system should detect whether a room is occupied and control its lights accordingly. This feature is intended to reduce unnecessary electricity consumption by ensuring that lights are automatically switched off when rooms are not in use.

2.3. Distributed Control Using ESP32 Microcontrollers

The system will consist of three control terminals: one for Lecture Room 1, another for Lecture Room 2, and a third for the main entrance. Each terminal may use a separate ESP32 microcontroller to manage its assigned functions.

The system design should ensure effective communication and coordination among the terminals, allowing the different components to operate reliably as part of one integrated solution.

2.4. Data Storage and Management

Given the potentially large volume of data generated by access logs, occupancy records, and system activities, management has proposed the use of a memory card module, such as an SD card module, to store additional information.

2.5. Serial Interface and Management Menus

Users should be able to access relevant system information through commands entered via the serial interface. The system designer is expected to develop suitable menus and commands that support efficient monitoring, configuration, and decision-making.

The proposed interface should provide useful information, such as access records, current building occupancy, room occupancy status, lighting status, and other relevant system information. Additional interfaces for outputting important data may be added.

2.6. Configurable Building Access Schedule

Management requires the ability to configure the time period during which building access is permitted. Outside the configured access period, the system must restrict access through the designated entry and exit doors.

The system must enforce the configured schedule consistently, including the requirement that individuals should not be permitted to leave the building outside the authorized period while they are still inside.

The access schedule must be adjustable at any time through an appropriate interface of the designer's choice. However, only authenticated and authorized personnel should be allowed to modify these settings.

2.7. Resource Synchronization and System Reliability

Since the system may involve multiple ESP32 microcontrollers accessing shared information and resources, appropriate synchronization mechanisms must be implemented. These mechanisms should prevent conflicts, inconsistent records, and race conditions when multiple operations occur concurrently.

The system should maintain accurate access records, occupancy information, and configuration settings while ensuring reliable coordination among the control terminals.

3. Assignment Task

Using the requirements outlined above, design and implement a Smart Control and Monitoring System for CoCIS using ESP32 microcontrollers and the Wokwi simulation platform.

The implementation should demonstrate automated access control, RFID-based identification, occupancy monitoring, intelligent lighting control, data storage, configurable access schedules, authenticated system configuration, and appropriate resource synchronization.

Students are encouraged to introduce innovative features that improve the system's security, efficiency, reliability, and usefulness to college management.

4. Required Deliverables

Each group must submit a report describing the design, implementation, and operation of the proposed system. The report must include the following:

1. **System Operation Report:** A detailed explanation of how the system works, including its architecture, hardware components, software design, and implemented features.
2. **Wokwi Simulation Link:** A link to the Wokwi project demonstrating the system implementation.
3. **Individual Video Demonstrations:** A link to a screen recording of each group member explaining the system and demonstrating the code sections or features they personally worked on.
4. **Individual Contributions:** A clear description of the tasks and responsibilities completed by each group member.
5. **Group Member Details:** The names and relevant identification details of all group members.

5. General Instructions

1. Students must work in groups consisting of no more than five members.
2. The coursework submission deadline is **2nd November 2026**.
3. Each group will be required to present its work to the lecturer.
4. Only students who attend the presentation will be awarded marks.
5. Use the WOKWI simulator and ESP-IDF
6. Submit the report on muele before the deadline

6. Assessment Criteria

Marks will be awarded based on the quality and completeness of the report, the ability of each student to answer questions during the presentation, and the clarity and effectiveness with which students explain their contributions in their recorded videos.

Particular emphasis will be placed on innovation and the additional features introduced to improve the proposed system.

Students must demonstrate a clear understanding of the features and code they claim to have implemented. Marks will be deducted if a student is unable to explain or answer questions about their stated contributions.

All group members are therefore expected to participate actively in the design, implementation, documentation, and presentation of the system.